⚙️ Configuration

Provider: `OPENAI`

Model: `gpt-4o-mini`

Max Budget Cap: `$0.10`

Timeout: `10.0s`

Preset Demo Queries

Select a sample query:

Custom Query

🔍 Autonomous AI Research Agent

Predixion AI Platform Engineering Internship Demonstration

Enter Research Question:

Compare LangChain, LlamaIndex, and Haystack for building production-ready RAG applications. Evaluate architecture, integrations, scalability, limitations, and developer experience.

🚀 ...

Researching: **Compare LangChain, LlamaIndex, and Haystack for building production-ready RAG applications. Evaluate architecture, integrations, scalability, limitations, and developer experience. Prefer official documentation and provide source citations for every major finding.**

✓ ✅ Research Workflow Complete!

Executive Short Answer

The comparison of LangChain and Haystack reveals strengths in architecture and integrations, while LlamaIndex lacks sufficient evidence for a thorough evaluation. LangChain offers a modular architecture suitable for various applications, whereas Haystack excels in integration with vector databases. However, limitations in evidence for LlamaIndex hinder a complete assessment.

Confidence L...	Verified Sources	Execution Time	Tokens Used
LOW	2	37....	8,3...

↑ Est: $0.0...

Confidence Analysis: LOW (Score: 71.2/100): Limited coverage (30%), 10 source(s) with partial verification.

<details>
<summary>📊 Evidence Coverage Matrix across Entities & Dimensions</summary>

Coverage Ratio: 30% | **Covered:** 1/18 dimensions

	LangChain	LlamaIndex	Haystack
Architecture	COVERED	MISSING	PARTIAL
Scalability	PARTIAL	MISSING	PARTIAL
Ecosystem	PARTIAL	MISSING	PARTIAL
Limitations	MISSING	MISSING	MISSING
Performance	PARTIAL	MISSING	PARTIAL
Database Integration	PARTIAL	MISSING	PARTIAL

</details>

💡 Key Findings

1. LangChain features a modular architecture that supports various integrations, including machine learning and IoT, as demonstrated in a predictive maintenance project that utilizes LangChain agents for decision-making and operator interaction.

2. Haystack is noted for its strong integration capabilities with vector databases like Milvus, which enhances its scalability and performance in RAG applications, allowing for efficient similarity searches and data management.

📚 Verified Sources Used

<details>
<summary>🔗 LangChain-Driven Predictive Maintenance for Industrial IoT [OFFICIAL_REPO] ⭐ (Official)</summary>

URL: https://github.com/jalsulaiman/LangChain-Driven-Predictive-Maintenance-for-Industrial-IoT-Overview

Relevance: This source provides insights into LangChain's architecture and its application in a real-world predictive maintenance system, showcasing its modular capabilities and integration with IoT.

</details>

Milvus (vector database) [COMMUNITY_ARTICLE]

URL:
https://en.wikipedia.org/wiki/Milvus_%28vector_database%29

Relevance: This source discusses the integration of Haystack with Milvus, highlighting its scalability and performance advantages in managing large-scale vector data for RAG applications.

Source Authority Breakdown

OFFICIAL_REPO (1)

- GitHub - jalsulaiman/LangChain-Driven-Predictive-Maintenance-for-Industrial-IoT-Overview: Showcase of LangChain-driven predictive maintenance system for Industrial IoT. Includes project overview, architecture diagrams, and demo outputs. · GitHub (https://github.com/jalsulaiman/LangChain-Driven-Predictive-Maintenance-for-Industrial-IoT-Overview)

ACADEMIC_PAPER (3)

- [2303.07991v1] Finding the Needle in a Haystack: Unsupervised Rationale Extraction from Long Text Classifiers (http://arxiv.org/abs/2303.07991v1)
- [2006.04533v1] Investigation Into the Viability of Neural Networks as a Means for Anomaly Detection in Experiments Like Atlas at the LHC (http://arxiv.org/abs/2006.04533v1)
- [2601.13383v1] A Lightweight Modular Framework for Constructing Autonomous Agents Driven by Large Language Models: Design, Implementation, and Applications in AgentForge (http://arxiv.org/abs/2601.13383v1)

COMMUNITY_ARTICLE (4)

- Milvus (vector database) (https://en.wikipedia.org/wiki/Milvus_%28vector_database%29)

- Object storage (https://en.wikipedia.org/wiki/Object_storage)
- Trusted execution environment (https://en.wikipedia.org/wiki/Trusted_execution_environment)
- Milvus (vector database) - Wikipedia (https://en.wikipedia.org/wiki/Milvus_%28vector_database%29)

GENERAL_REPO (2)

- multi-agent-code-reviewer - Ranjan-10000/multi-agent-code-reviewer (https://github.com/Ranjan-10000/multi-agent-code-reviewer)
- WhisperingChunks-Overview - VinodSridharan/WhisperingChunks-Overview (https://github.com/VinodSridharan/WhisperingChunks-Overview)

📌 Key Assumptions

- Information is current based on publicly available documentation and technical publications indexed up to present.
- Benchmark and performance assertions assume standard production hardware environments unless otherwise noted.

⚠️ Limitations

- Insufficient independent evidence gathered for entity: 'LlamaIndex'.
- Comparative evidence for evaluation dimension 'Limitations' is incomplete.
- Limited or fragmented supporting evidence; cross-verification across independent sources was weak.

🎯 Suggested Next Steps

- Conduct further research to gather more comprehensive evidence on LlamaIndex.

- Explore user experiences and case studies for LangChain and Haystack to better understand developer experience and limitations.

<details>
<summary>🔬 Agent Orchestration Trace (Observability & Retrieval Rounds)</summary>

Retrieval Rounds: 2

Second-Hop Queries: ['LangChain scalability documentation benchmarks', 'LangChain ecosystem documentation benchmarks']


```
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  "To systematically compare LangChain, LlamaIndex, and Haystack based on specified evaluation dimensions for building production-ready RAG applications."
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    "Evaluate the architecture of LangChain, LlamaIndex, and Haystack."
    1 :
    "Investigate integrations available for LangChain, LlamaIndex, and Haystack."
    2 :
    "Assess the scalability of LangChain, LlamaIndex, and Haystack."
    3 :
    "Identify limitations of LangChain, LlamaIndex, and Haystack."
    4 :
    "Evaluate the developer experience with LangChain, LlamaIndex, and Haystack."
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    0 : "web_search"
    1 : "webpage_extractor"
  ]
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    0 : "LangChain architecture overview"
    1 : "LlamaIndex architecture details"
```

</details>

```
    2 :
    "Haystack architecture documentation"
    3 :
    "LangChain scalability documentation
    benchmarks"
    4 :
    "LangChain ecosystem documentation be
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"pages_extracted_count" : 5
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    1 : "LlamaIndex architecture details"
    2 :
    "Haystack architecture documentation"
]
▼ "round_2_queries" : [
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    1 :
    "LangChain ecosystem documentation be
    nchmarks"
]
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    "retries_count" : 0
    "fallback_calls_count" : 0
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}
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Raw Validated JSON Output (Pydantic Schema)

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    "Haystack is noted for its strong integra
  ],
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      "relevance": "This source provides inst
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    "Limited or fragmented supporting evidenc
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    "Explore user experiences and case studie
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```

```
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      [
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```

```
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        "Limitations": "MISSING",
        "Performance": "MISSING",
        "Database Integration": "MISSING"
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        "Database Integration": "PARTIAL"
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  },
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    ],
    "ACADEMIC_PAPER": [
      "[2303.07991v1] Finding the Needle in a
      "[2006.04533v1] Investigation Into the
      "[2601.13383v1] A Lightweight Modular F
    ],
    "COMMUNITY_ARTICLE": [
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      "Object storage (https://en.wikipedia.o
      "Trusted execution environment (https:/
      "Milvus (vector database) - Wikipedia (
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      "multi-agent-code-reviewer - Ranjan-100
```

```
      "WhisperingChunks-Overview - VinodSridh
    ]
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        "http://arxiv.org/abs/2601.13383v1"
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    }
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        "Ecosystem": "PARTIAL",
        "Limitations": "MISSING",
        "Performance": "PARTIAL",
        "Database Integration": "PARTIAL"
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```

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          "Ecosystem": "PARTIAL",
          "Limitations": "MISSING",
          "Performance": "PARTIAL",
          "Database Integration": "PARTIAL"
        }
      }
    }
}
```